Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		<u>Flux</u> in heated object varies (1) Faraday's law referred to or implied (1) Current flows, [providing heating] (1) High frequency increases rate of change of flux (1)	1	1 1 1		4		
	(b)		**Indicative content:** **Diagram** Diagram of Hall probe / chip / wafer B-field shown Current flow shown perpendicular Force direction shown correctly Voltmeter correctly connected Slice orientated / adjusted in field **Explanation and measuring B** Electrons / charge carriers experience force Given by FLHR They move to one side Setting up Hall voltage or field Equilibrium is reached between magnetic and electric forces Hall voltage is proportional to B **OR** equation quoted Can be calibrated to give B directly **OR** hence B calculated **5-6 marks** Comprehensive diagram and explanation and measuring $\boldsymbol{B}$. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.*	6			6		6

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			3-4 marks Comprehensive diagram **or** explanation and measuring ***B*** **or** limited attempt at both. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.* **1-2 marks** Limited attempt at either the diagram **or** explanation and measuring ***B*.** *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** No attempt made or no response worthy of credit.						
			Question 5 total	**7**	**3**	**0**	**10**	**0**	**6**